

Question

Element **X** can react with AgF under strictly dry and low-pressure conditions to obtain an isomer **A**₁ of compound **A** with C_2 symmetry. Another isomer **A**₂ can be obtained by reacting KSO_3F with **B** in liquid SO_2 . **A**₂ is thermodynamically unstable and can disproportionate into compound **C** and elemental **X** under HF catalysis. **C** can form a 1:1 adduct **D** with CsF . Chlorination of **B** in the presence of FeCl_3 yields compound **E**. Reacting **E** with NaF in warm diethyl ether yields a mixture of **B** and **C**. Both **B** and **E** can react with NH_4Cl to produce a cyclic compound **F**. **F** is insoluble in water but soluble in concentrated NaOH solution to form two sodium salts **G** and **H**. The mass fractions of Na in **G** and **H** are 29.09 % and 36.48 %, respectively. It is known that all substances represented by letters in the question contain **X**. From the following options, choose the one that best fits the meaning of the question.

- A.** All other options are incorrect
- B.** **A**₁ and **A**₂ have different structures, but both possess a diad axis.
- C.** **A**₂ and **B** have different structures, but their symmetry elements are completely identical.
- D.** The structure of **C** and the corresponding structure of the substance in **D** that forms a 1:1 adduct, although slightly different, share the same hybridization mode of the central atom.
- E.** The structure of **C** and the corresponding structure of the substance in **D** that forms a 1:1 adduct, although slightly different in the hybridization mode of the central atom, share the same symmetry elements.
- F.** The valence states of **C**, **E** are consistent with those of the cyclic compound **F** formed.
- G.** The valence states of **C** and **E**, as well as the valence state of the formed cyclic compound **F**, are all distinct from each other.

Answer

Correct Answer: **G**

Detailed Explanation

The breakthrough point of the problem lies in the information of compounds **G** and **H**.

Calculating the molar mass of **G** and **H**:

For **G**: The mass fraction of sodium is 29.09%.

The molar mass of **G** satisfies:

If $n = 1$, $M(\mathbf{G}) \approx 79.03\text{g/mol}$, the mass of the anion part $\approx 79.03 - 22.99 = 56.04\text{g/mol}$.

If $n = 2$, $M(\mathbf{G}) \approx 158.06\text{g/mol}$, the mass of the anion part $\approx 158.06 - 45.98 = 112.08\text{g/mol}$.

If $n = 3$, $M(\mathbf{G}) \approx 237.09\text{g/mol}$, the mass of the anion part $\approx 237.09 - 68.97 = 168.12\text{g/mol}$.

Because it is treated in concentrated NaOH solution, and the anion is composed of elements **X** and O, **G** is $\text{Na}_2\text{S}_2\text{O}_3$.

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G is $\text{Na}_2\text{S}_2\text{O}_3$

Similarly, for **H**: The mass fraction of sodium is 36.48%, calculated to be **H** is Na_2SO_3 .

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H is Na_2SO_3

Therefore, element **X** is S.

F is hydrolyzed in NaOH to obtain Na_2SO_3 and $\text{Na}_2\text{S}_2\text{O}_3$. We know that **F** is a cyclic compound, so we can deduce that **F** is cyclic S_4N_4 .

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F is cyclic S_4N_4

B or **E** can react with NH_4Cl to obtain S_4N_4 ; **B** is chlorinated in the presence of FeCl_3 to obtain **E**, one of which is S_2Cl_2 , and the other should be SCl_2 . The latter has a higher valence, so **B** = S_2Cl_2 , **E** = SCl_2 .

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B = S_2Cl_2

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E = SCl_2

Elemental sulfur reacts with AgF to obtain **A**, whose isomers **A**₁ belong to the C_2 point group and **A**₂ belong to the C_s point group, that is, **A**₁ = F-S-S-F, **A**₂ = S=SF₂. It can be known that the symmetry elements of **A**₂ and **A**₁, **B** are different.

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A is S₂F₂

CHECKPOINT

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A₁ is F-S-S-F, belonging to the C_2 point group

CHECKPOINT

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A₂ = S=SF₂, belonging to the C_s point group

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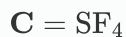
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The symmetry elements of **A**₂ and **A**₁, **B** are different

A₂ disproportionates to produce **C** and S, so **C** = SF₄, and the hybridization is sp^3d hybridization.

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According to the above information, we can know that the valences of **C**, **E** and the valence of the cyclic compound **F** formed are all different from each other.

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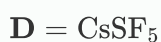
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The valences of **C**, **E** and the valence of the cyclic compound **F** formed are all different from each other.

C reacts with CsF to form a 1 : 1 adduct **D**, that is, $\mathbf{D} = \text{CsSF}_5$, and the hybridization is sp^3d^2 hybridization.

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CHECKPOINT

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The hybridization modes of S atoms in **C** and **D** are different, which are sp^3d hybridization and sp^3d^2 hybridization, respectively

CHECKPOINT

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The anion symmetry of **C** and **D** are different